

Why do people smoke ?

Rinky Abraham

Data Science Trainee
Springboard

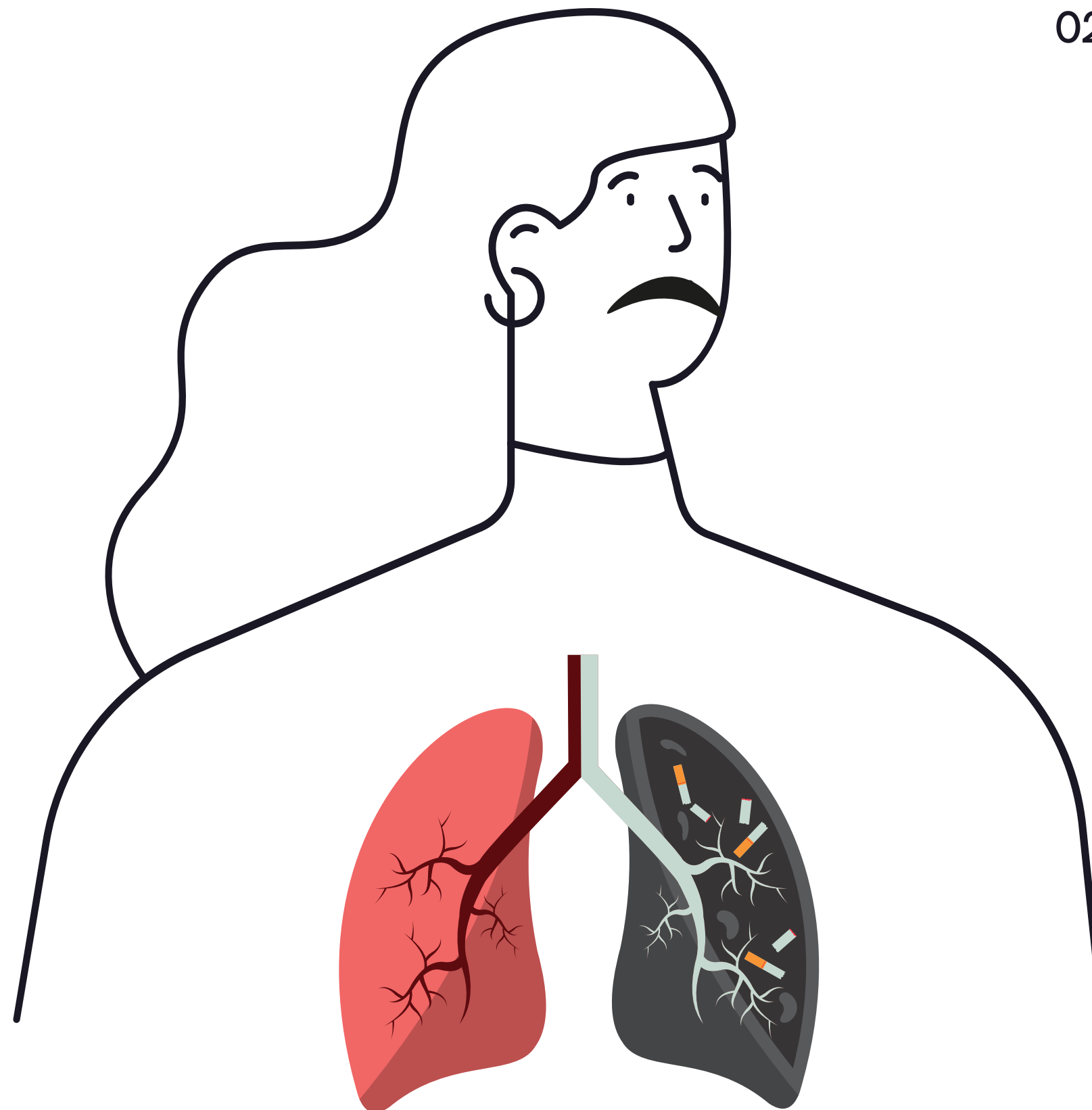


Why do we care ?

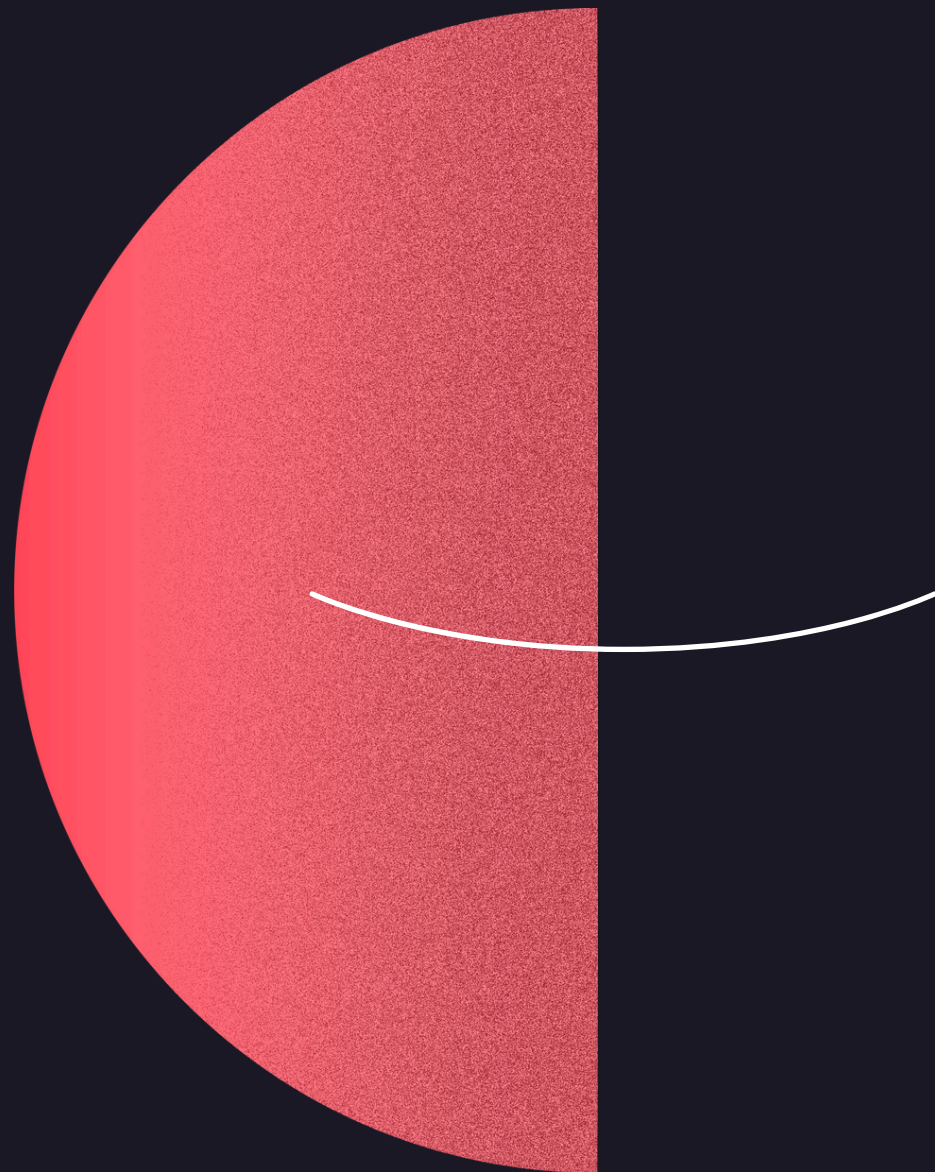
02

IN 2023,
GLOBALLY **~7 MILLION** LIVES WERE LOST
USA **~500,000** LIVES

POLICY INITIATIVES ARE EVERYWHERE , BUT
ARE THEY REALLY **WORKING ??**



What we want to answer ?



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What factors (defined by Health Belief Model) (HBM) contribute most towards the habit of smoking ? Are there any policy implications?

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Does this trend vary by age group ?

DATA

04

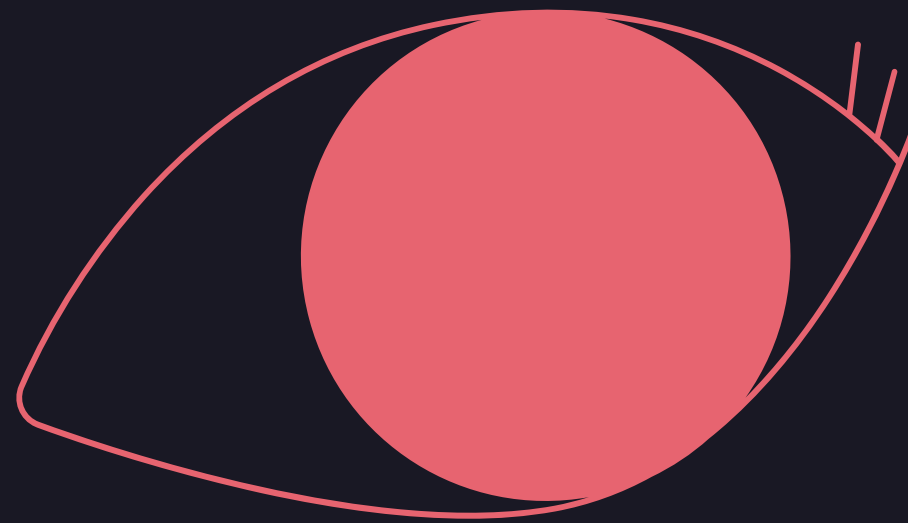
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DATASET: BEHAVIORAL RISK
FACTOR SURVEILLANCE SYSTEM
(BRFSS)

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SOURCE: KAGGLE API

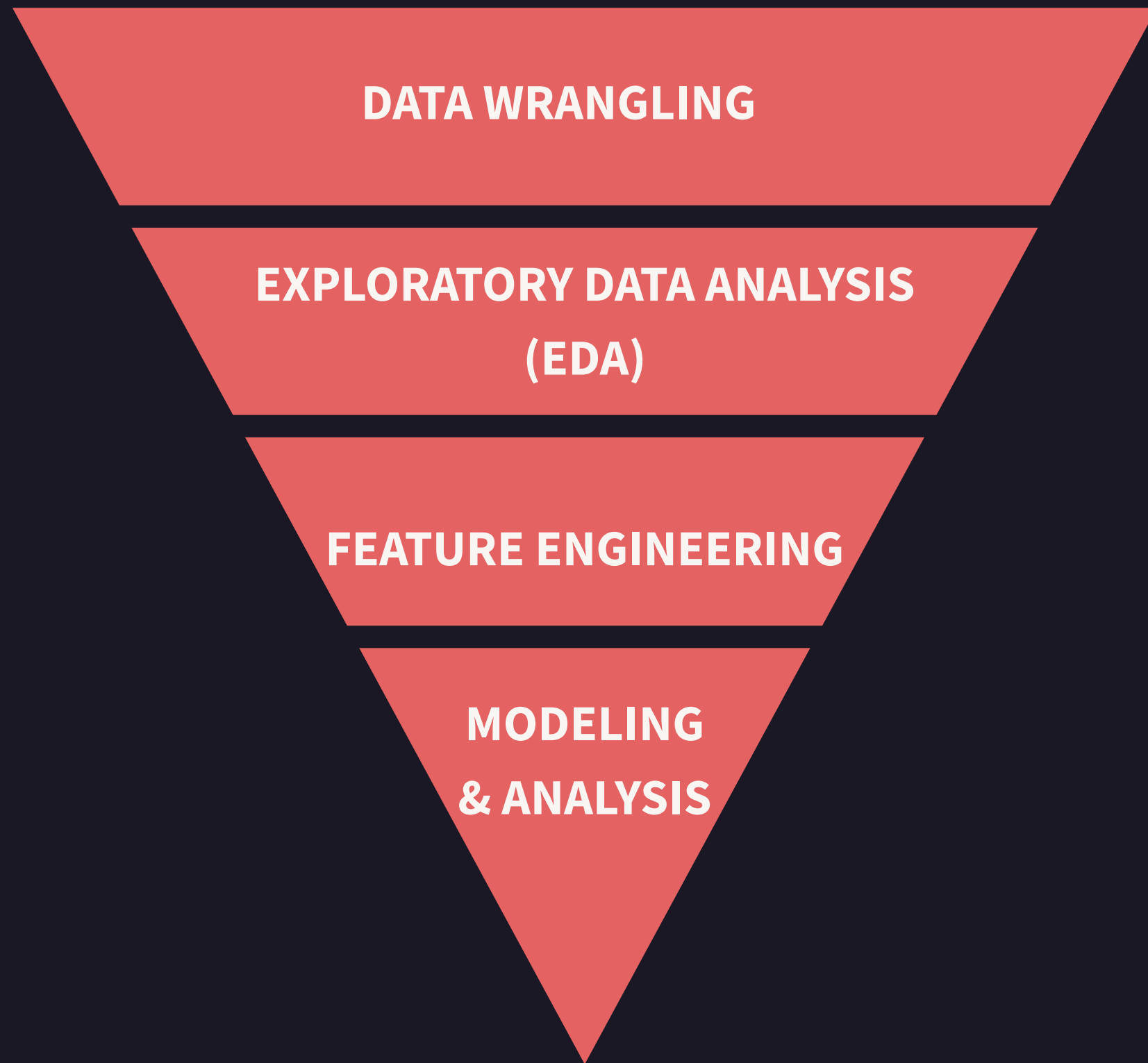
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SIZE: ~ 2MILLION RECORDS AND
645 COLUMNS

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DATA TYPE: SURVEY





METHODOLOGY



SMOKER PREDICTION MODEL

HEALTH BELIEF MODEL

Widely used frameworks in public health (since 1950s) for understanding why people do or don't take certain health action





DATA WRANGLING

MISSING DATA

Drop columns with
missingness > 0.75

```
threshold =  
len(combined_df)*.75  
df.dropna(axis=1,  
thresh=threshold)
```

645 to 87 columns

NON-INFORMATIVE FEATURES

Drop redundant
features

```
df.drop(['SEQNO',  
'CTELENUM'],  
axis=1)
```

87 to 84 columns

DUPLICATE FEATURES

Drop record
identifiers

```
df.drop(['INCOME2'],  
axis=1)
```

84 to 58 columns

WRONG DATATYPES

Change datatypes
appropriately

```
_EDUCAG and _INCOME  
Ordinal variables  
  
PHYSHLTH and  
MENTLHLTH  
Numeric variables
```

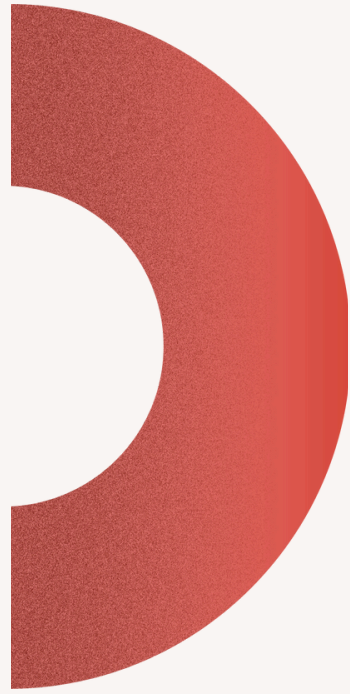
No Change

HANDLING ERROR VALUES & IMPUTE FEATURE VALUES

Standardize
feature values to
same units

```
Drop _IYEAR,  
_IMONTH, _IDATE  
  
Impute _HEIGHT2,  
_WEIGHT3 with mean
```

58 to 55 columns



EDA

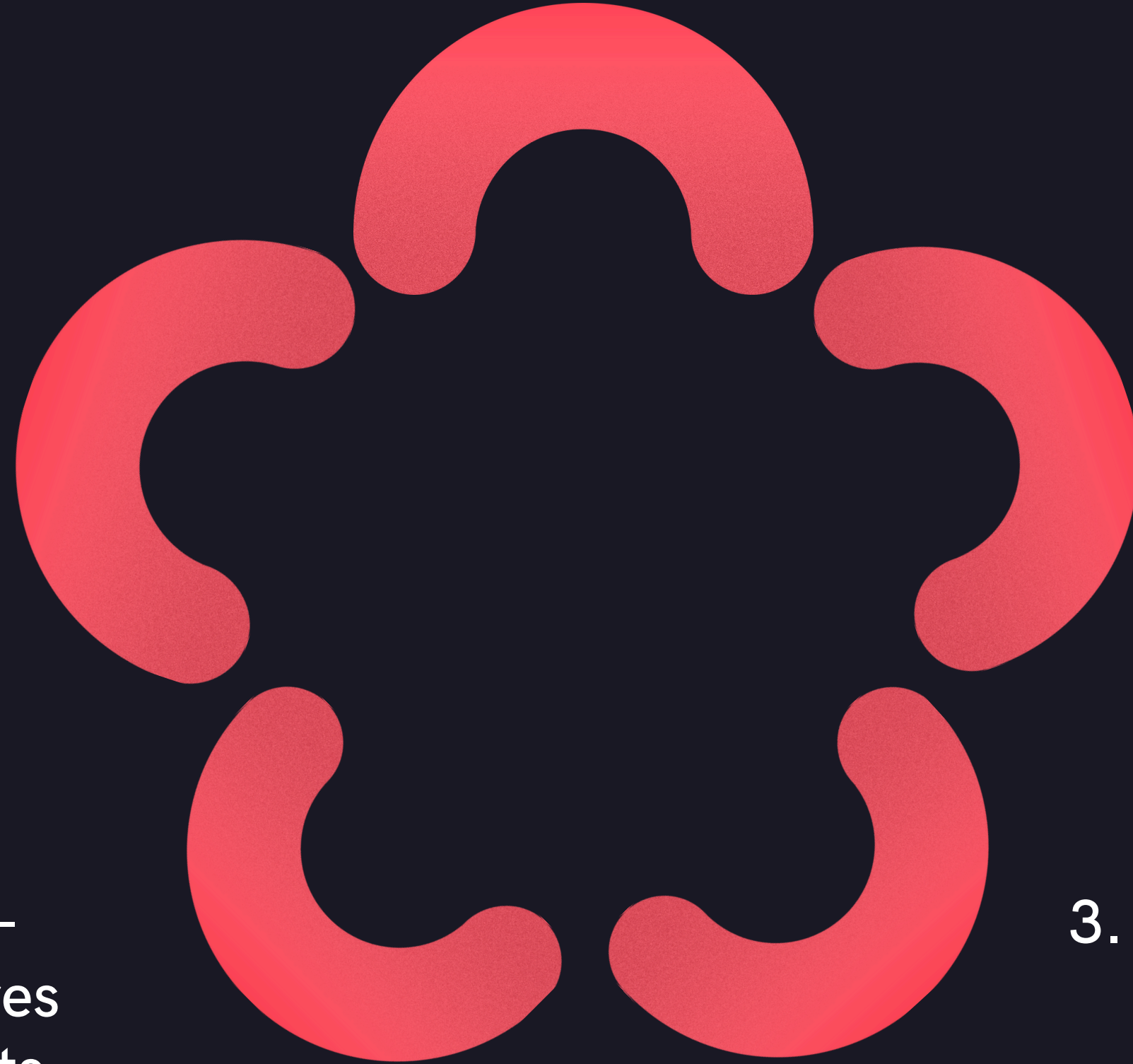
1. Selection of Target Variable

2. Data Validation

3. Map dataset
to HBM
constructs

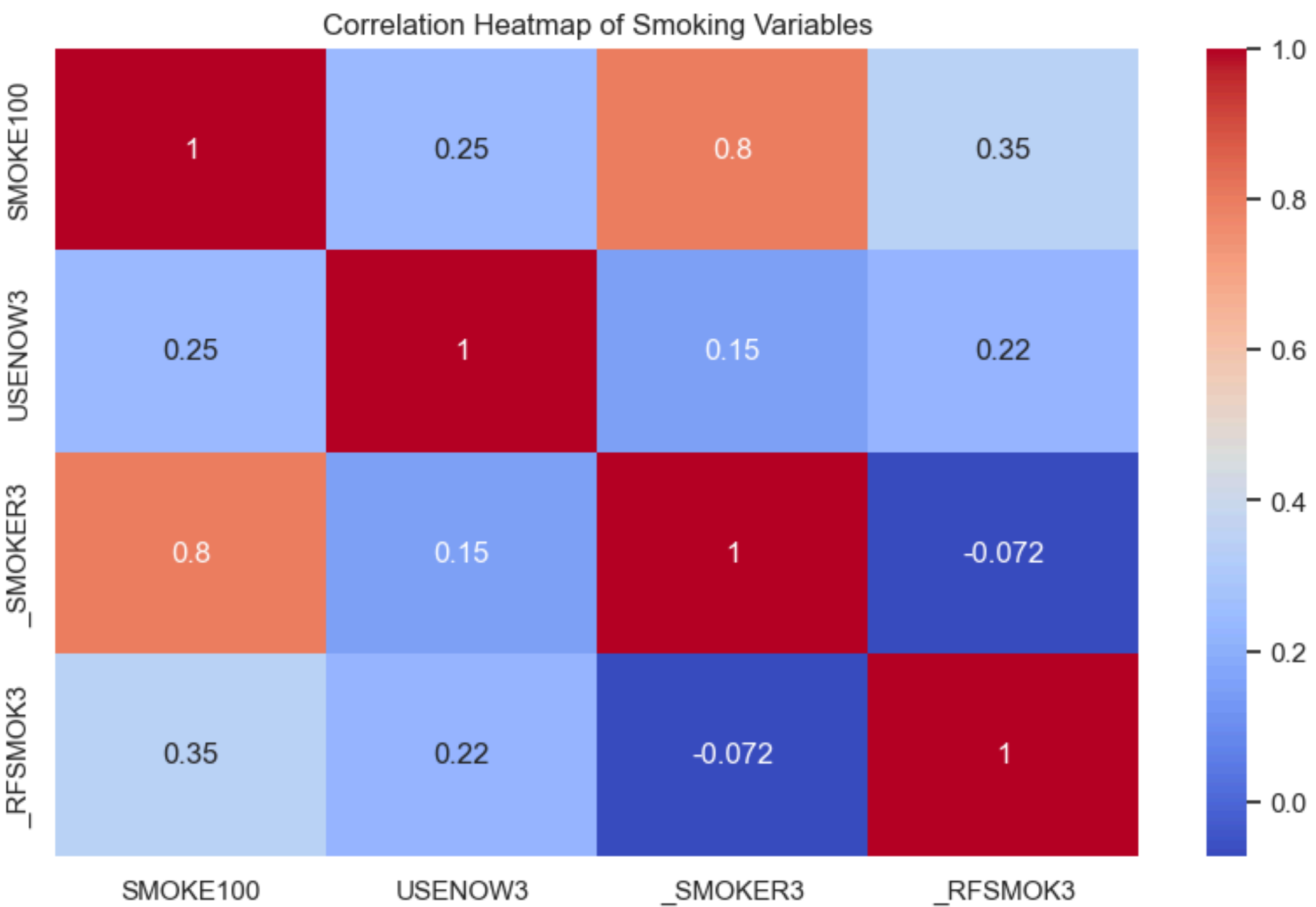
4. Binning zero-
imbalance features
into fewer buckets

5. Evaluating Data
Leakage



Selection of Target Variable

TARGET VARIABLE	DEFINITION	DISTINCT CLASSES
SMOKE100	Smoked at Least 100 Cigarettes	4 (Yes, No, Don't know, Refused)
USENOW3	Use of Smokeless Tobacco Products	5 (Every day, Some days Not at all, Don't know,Refused)
_SMOKER3	Computed Smoking Status	5 (Everyday smoker, Someday smoker, Former smoker, Non-smoker)
_RFSMOK3	Current Smoking Calculated Variable	3 (No, Yes, Don't know)



Map dataset to HBM constructs

Variable	HBM Construct(s)	Explanation
ASTHMA3	Susceptibility	Having asthma could indicate if the person is likely to get risks related to smoking
PHYSHLTH	Severity	number of days of poor physical health indicates how serious the risks of smoking are for the person
HLTHPLN1	Benefits	Having a health insurance helps reassure that the risks related to smoking is reduced
MEDCOST	Barriers	Medical cost; barrier to access care/cessation support which could affect how one cares for ones health ie. might increase smoking
CHECKUP1	Cues to Action	Routine checkup; triggers cues and reinforces benefits.
EXERANY2	Self- Efficacy	If the person has a self motivation to stay healthy by prioritising exercise, he has better motivation to continue taking healthy actions.
MARITAL	Other Variables	Socioeconomic factor that determines certain behaviour

Evaluating Data Leakage

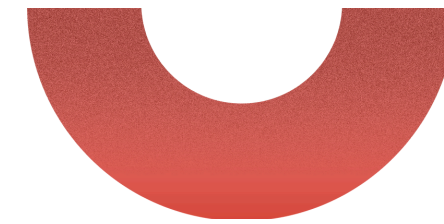
Chi-square test: To check linear relationship between target variable and each feature.

Cramer's V evaluates strength of relationship

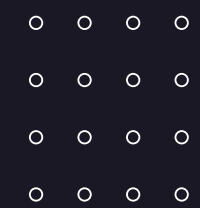
Observation: *Socioeconomic factors (education, income, marital status) have more straightforward linear relationships*

Mutual Information test: Measures non-linear relationships and information content shared between variables

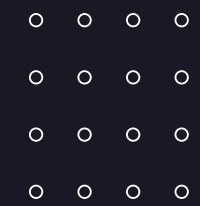
Observation: *Medical conditions (stroke, kidney disease, heart attack) have complex, non-linear relationships with your target that Chi-Square missed.*



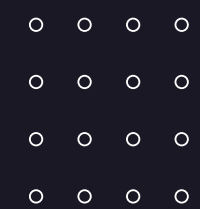
FEATURE ENGINEERING



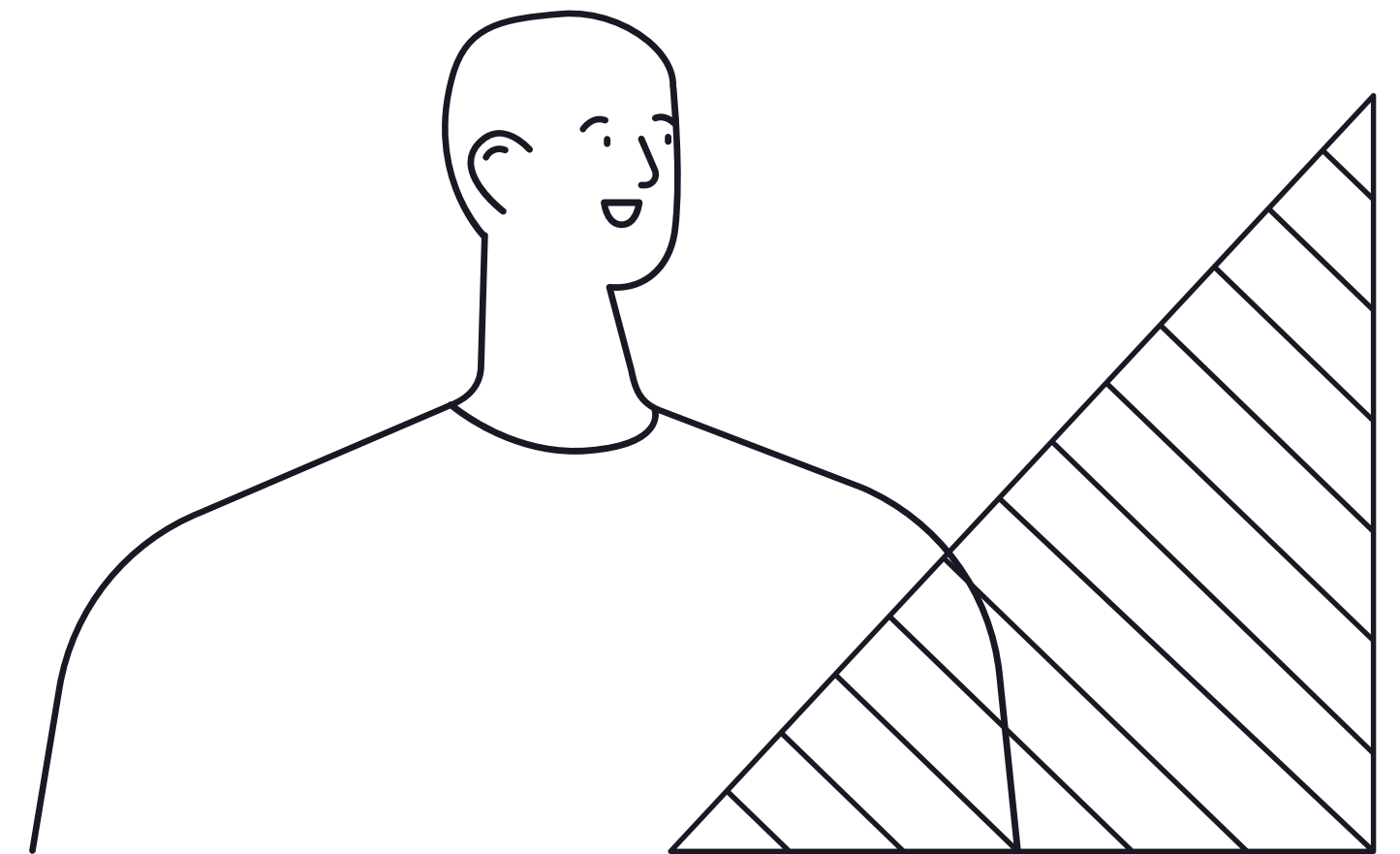
CATEGORICAL VARIABLES:
One-hot Encode

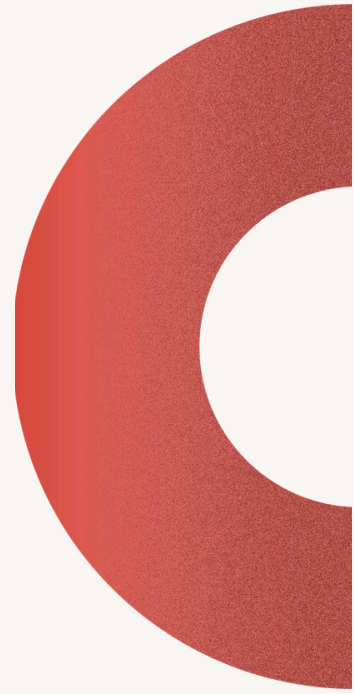


NUMERIC VARIABLES:
Standardize



GENERATE TRAIN/TEST SPLIT:
~1.9 million records for Train set
~500,000 records for Test set





MODELING

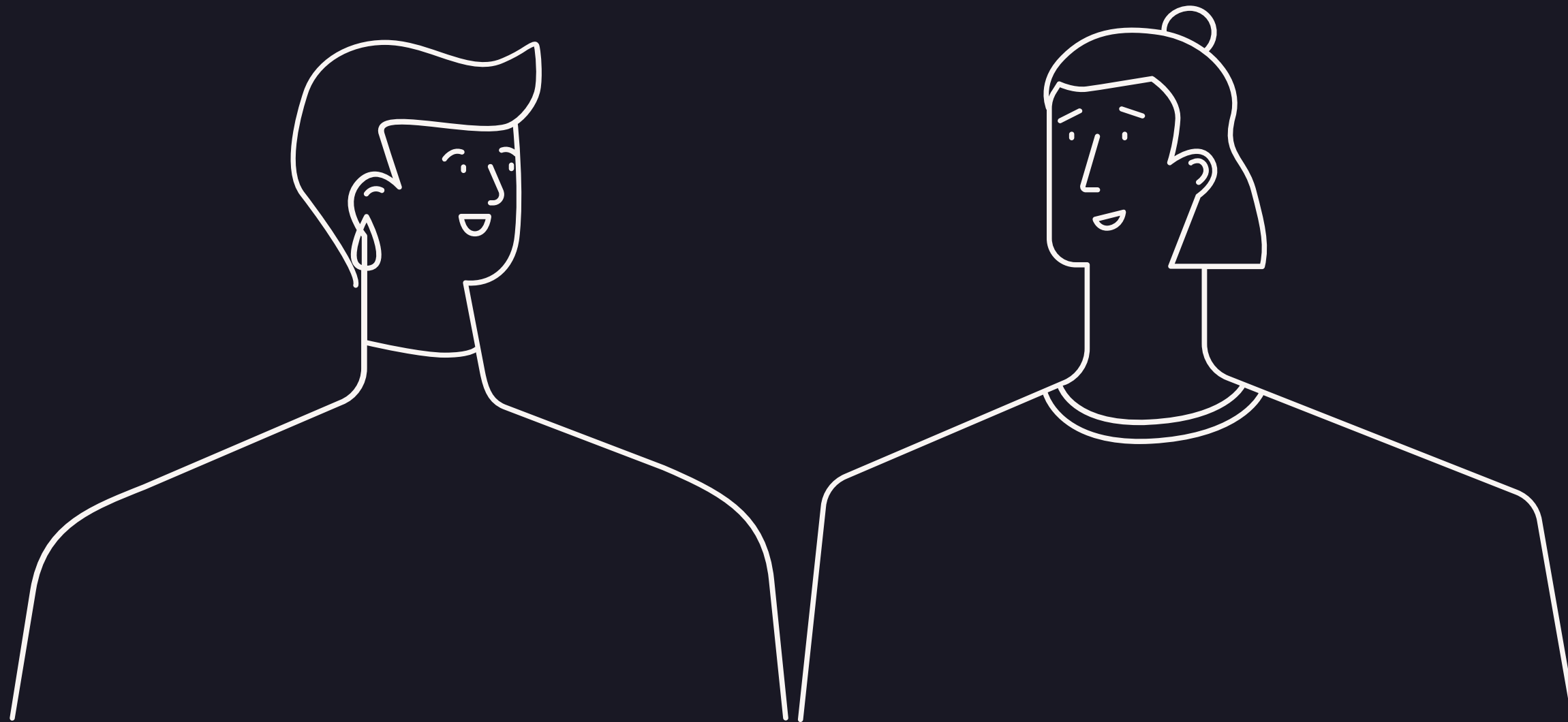
Model	Setup	Imbalance Handling	Key Params
CatBoost Default	Minimal, fast baseline	auto_class_weights='Balanced'	Defaults
CatBoost Tuned	Optimized for imbalance	Same as default	iterations=1000, depth=8, learning_rate=0.1, l2_leaf_reg=3, params
LightGBM Default	Basic config	class_weight='balanced'	num_leaves=31, lr=0.1, n_estimators=1000
LightGBM Tuned	RandomizedSearchCV tuned	Same as default	param_dist = { 'num_leaves': [31, 63, 127], 'max_depth': [-1, 5, 10, 15], 'learning_rate': [0.01, 0.05, 0.1], 'min_child_samples': [5, 10, 20], 'min_split_gain': [0, 0.001], 'subsample': [0.7, 0.9, 1.0], 'colsample_bytree': [0.7, 0.9, 1.0], 'reg_alpha': [0, 0.1, 0.5], 'reg_lambda': [0, 0.1, 0.5] }

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC	Training Time
CatBoost	0.729721	0.333598	¹ 0.713685	² 0.454670	² 0.796675	² 0.458032	1144.751788
CatBoost Tuned	0.730858	0.334517	0.712342	0.455249	0.796907	0.458349	1831.023640
LightGBM	0.728925	0.332754	0.713301	0.453807	0.795863	0.456280	16.839819
LightGBM Tuned	0.730335	0.333869	0.711520	0.454481	0.796365	0.457688	3298.030455

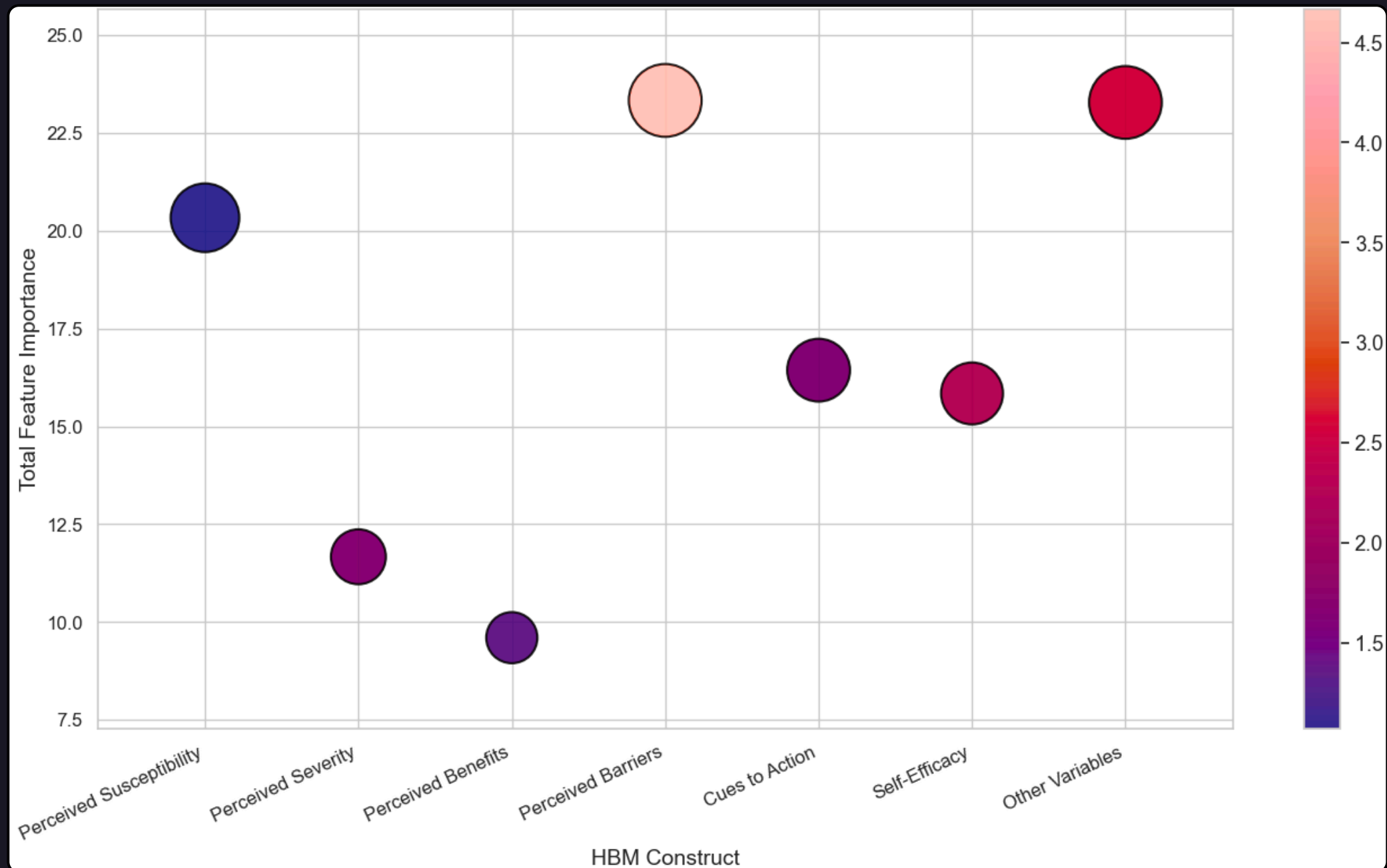
ANALYSIS

What constructs defined by Health Belief Model (HBM) contribute most towards the habit of smoking ? What are some policy implications ?

Does this trend vary by age group ?

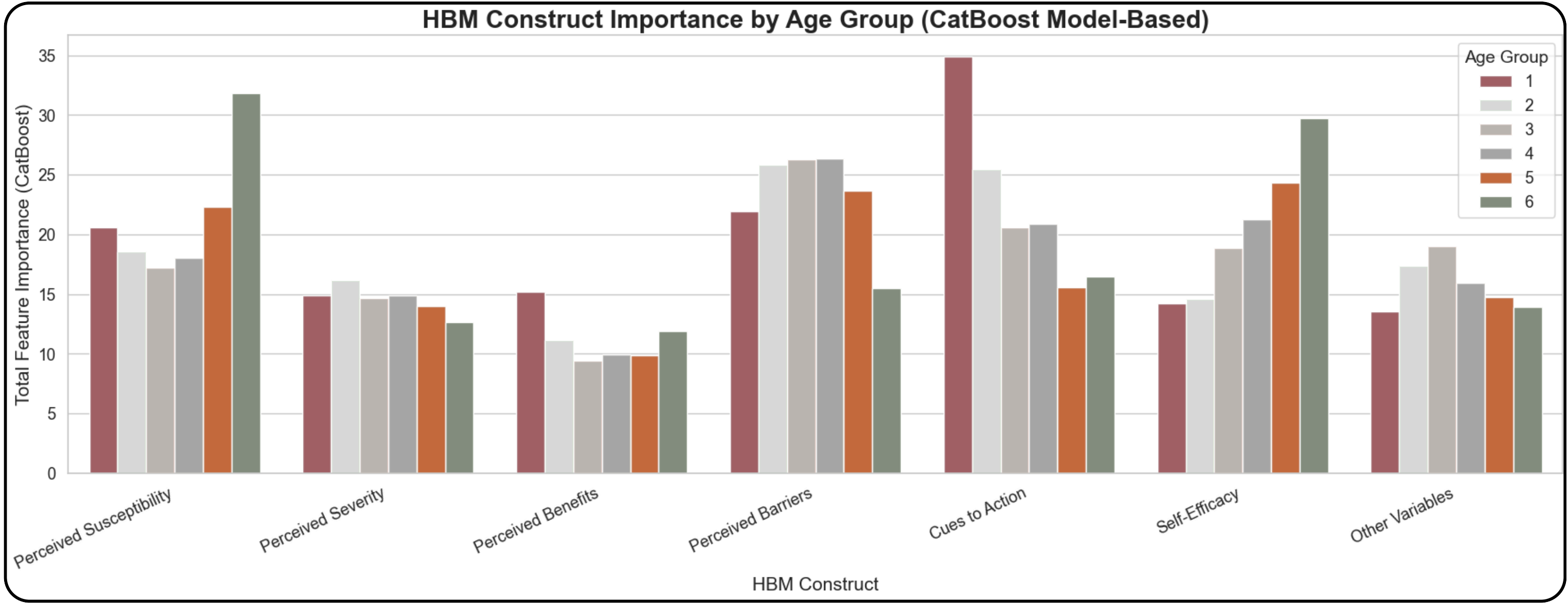


TOP PREDICTORS



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Feature	Importance (Prediction ValuesChange)	HBM Construct(s)
_EDUCAG	15.54	Barriers
_AGE65YR	9.62	Other
MARITAL	7.18	Other
_RFBING5	6.01	Cues to Action
WEIGHT2	5.06	Self-Efficacy
CHCCOPD1	4.89	Susceptibility
GENHLTH	4.8	Severity
_INCOMG	4.16	Barriers
HEIGHT3	3.47	Self-Efficacy



Code	Age Range	Description
1	18–24	Young Adults
2	25–34	Early Career Adults
3	35–44	Mid-Career Adults
4	45–54	Mature Adults
5	55–64	Pre-Retirees
6	65+	Seniors

18-24

Most influenced by **Cues to Action** and Perceived Benefits

Leverage peer influence, social media, benefits messaging.

25-54

Dominated by **Barriers**.

Address barriers: stress, time, cost, support access.

55-64

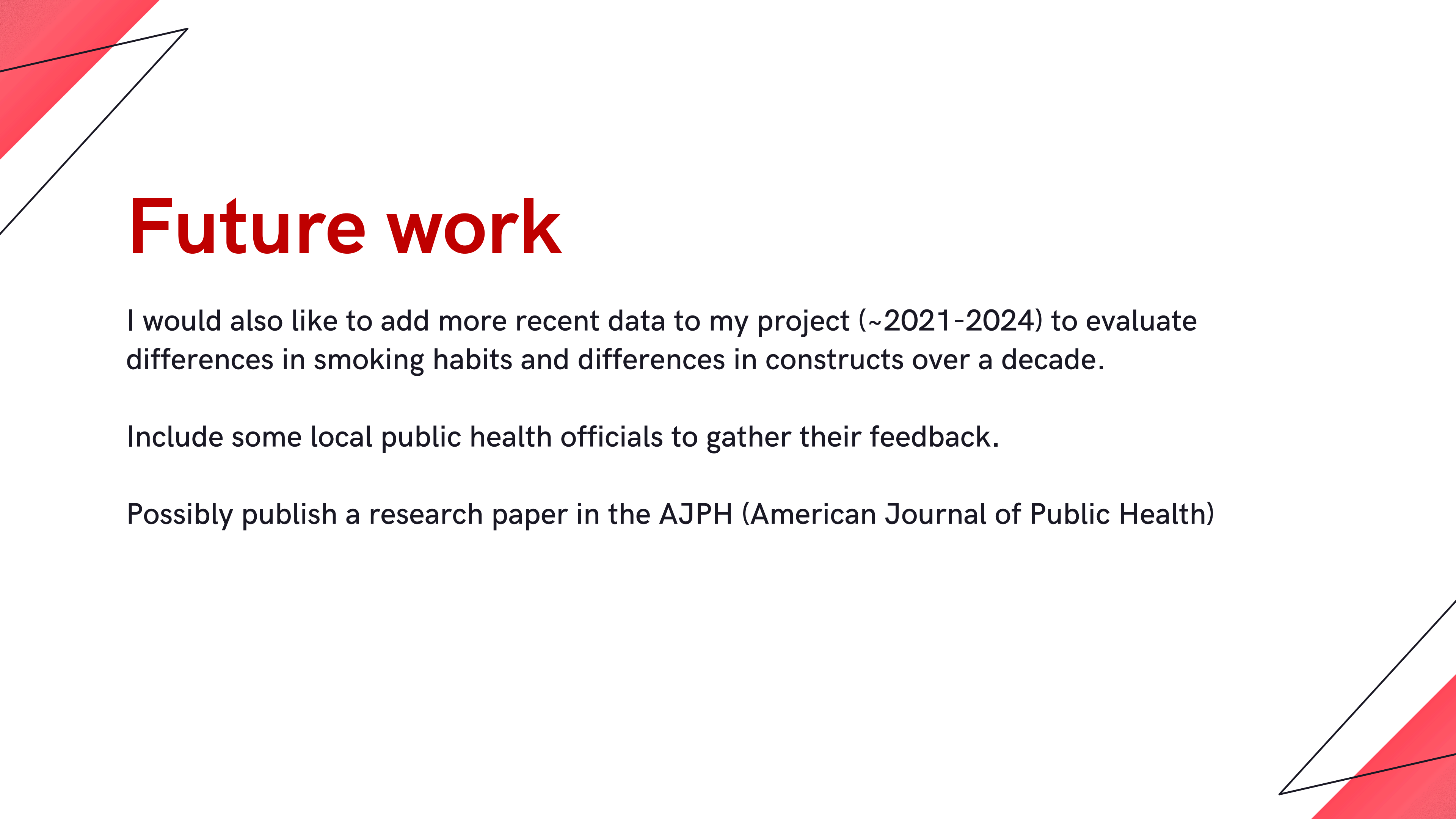
Influenced most by **Susceptibility**.

Emphasize risk and susceptibility via health condition links.

65+

High on **Self-Efficacy**, **Susceptibility**, and **Benefits**.

Empower with tools and reinforce confidence to quit.



Future work

I would also like to add more recent data to my project (~2021-2024) to evaluate differences in smoking habits and differences in constructs over a decade.

Include some local public health officials to gather their feedback.

Possibly publish a research paper in the AJPH (American Journal of Public Health)

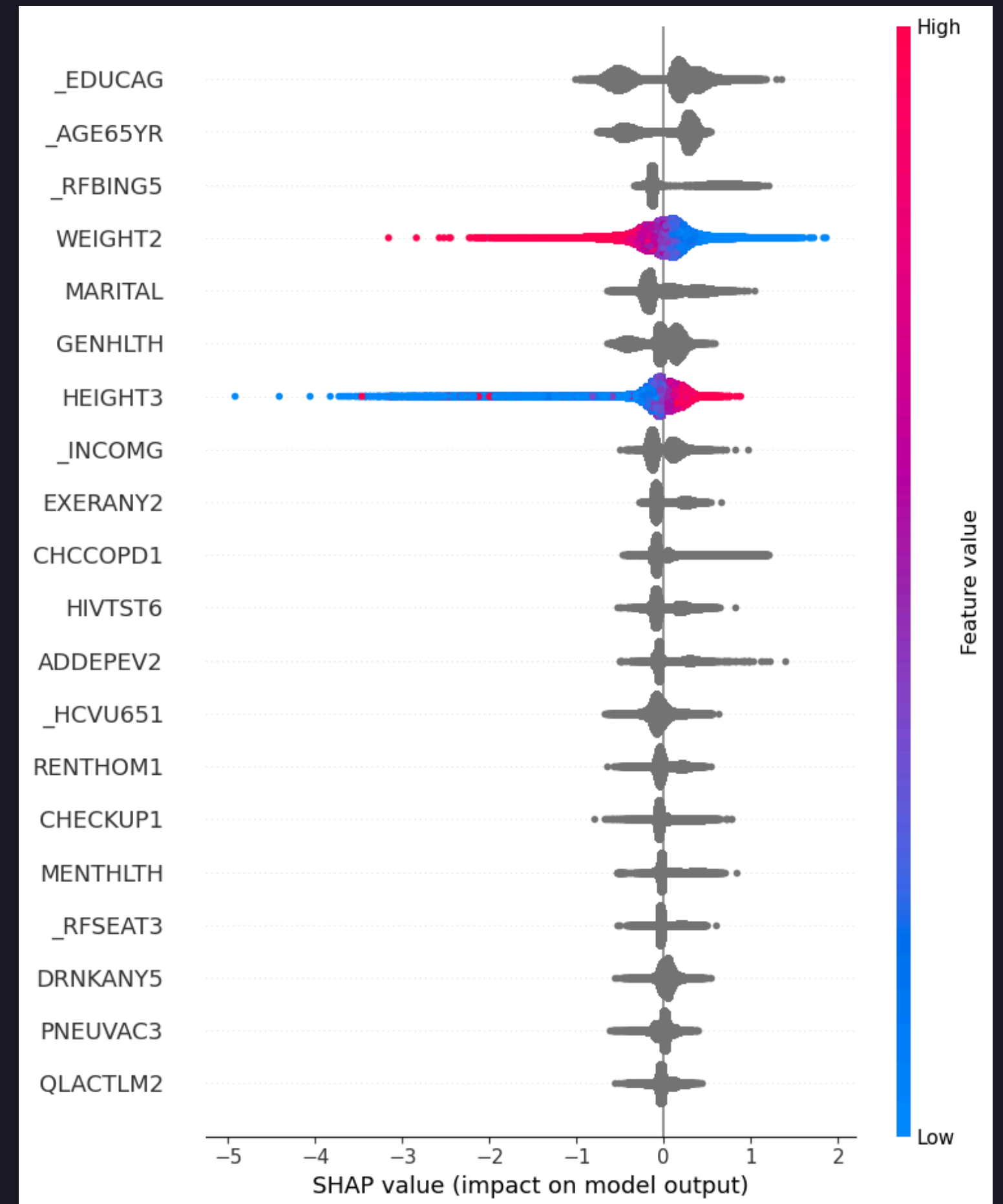
PART II -

Policy Recommendations

PRE-DATA WRANGLING

Pre-DataWrangling Steps to determine if we can use the current dataset to simulate various policy scenarios

1. Perform SHAP analysis on our prior model (CatBoost Model) to find out the most impactful levers guiding prediction of smoking
2. Perform SHAP analysis across age group



Outcomes

1. Our previous model has NaNs encoded either as 7 or 9, which needs to be handled to plot informative SHAP plots
2. Fix the order/encoding for some of the Ordinal variables to generate meaningful results on our SHAP plots
3. Label encode our variables to help in SHAP plots

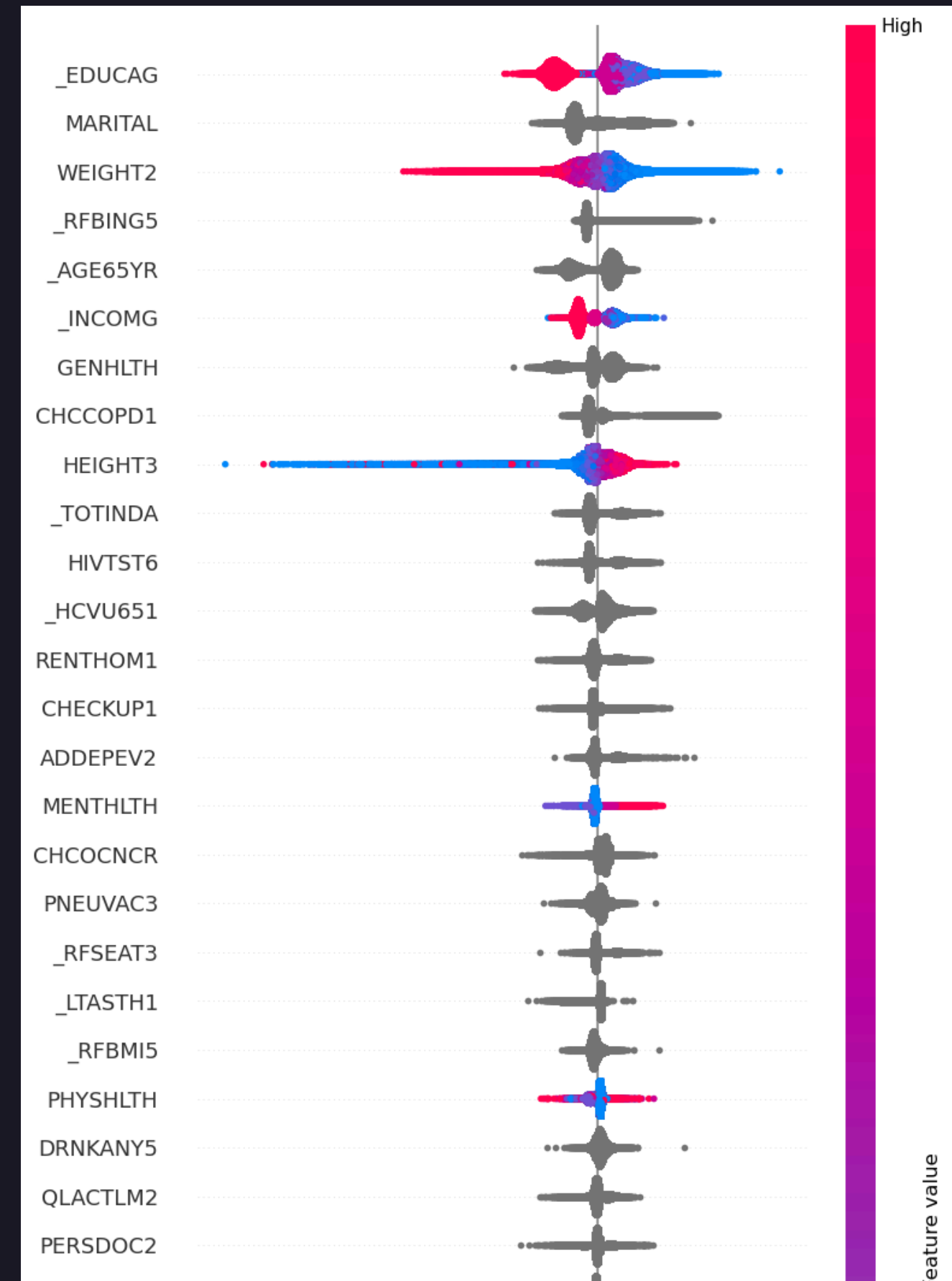
DATA WRANGLING

1. Large number of 7/9 values that don't give us any information.
2. Fix nominal ordering for categorical features like GENHLTH and _RFHLTH
3. Recode 7 to NaN, since Catboost Natively handles NaN values

EDA

Some tasks performed are -

1. Import new train/test data that has dealt with NaN
2. Import nominal/ordinal feature lists
3. Train new CatBoost Model for SHAP analysis on modified features
4. Train new CatBoost Model for SHAP analysis on modified features across age-groups



CONFOUNDER TEST

1. Check associations: 1.1. Each candidate variable $\leftrightarrow$ _INCOMG.
Each candidate variable $\leftrightarrow$ _RFSMOK3.
2. Compare regression models: 2.1. Crude: smoking ~ income.
Adjusted: smoking ~ income + candidate.

If the income coefficient changes $\geq 10\%$, candidate is a confounder.

Outcomes

The money-related variables that represent crucial levers and barriers for smokers include:

1. INCOME (_INCOMG) - Appears consistently across all age groups as a moderate to strong predictor, suggesting income level significantly impacts smoking behaviors/cessation success
2. Medical Costs (MEDCOST) - Shows up as a predictor across groups, indicating healthcare affordability affects smoking-related outcomes
3. Education (_EDUCAG) - While not directly money, education strongly correlates with earning potential and appears as a top predictor across all age groups
4. Healthcare Coverage (_HCVU651) - Insurance status appears as a factor, representing financial access to smoking cessation resources

Data Intervention Scenarios

Scenario A - Expand Mental Health Access

Goal: Reduce smoking by improving population mental health (MENTHLTH).

Rationale: MENTHLTH showed the largest confounding effect (up to ~28% change). Improving mental health should lower smoking propensity.

Intervention rule (synthetic):

For people in lowest two income groups (_INCOMG levels 1 & 2), reduce MENTHLTH measure by 20–30% (e.g. move categorical mental-health scores one level towards "better").

We shift 30% of respondents coded poor → fair, fair→good (stochastic selection within target group).

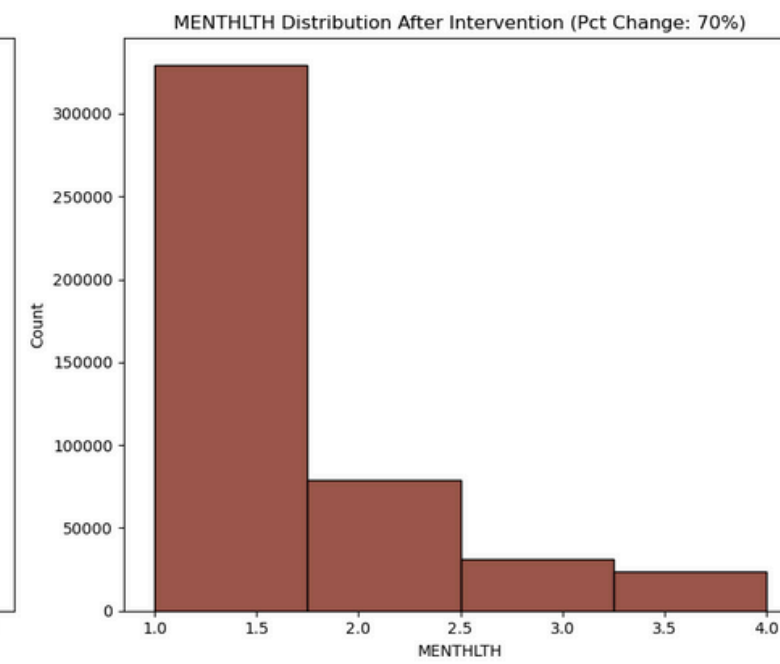
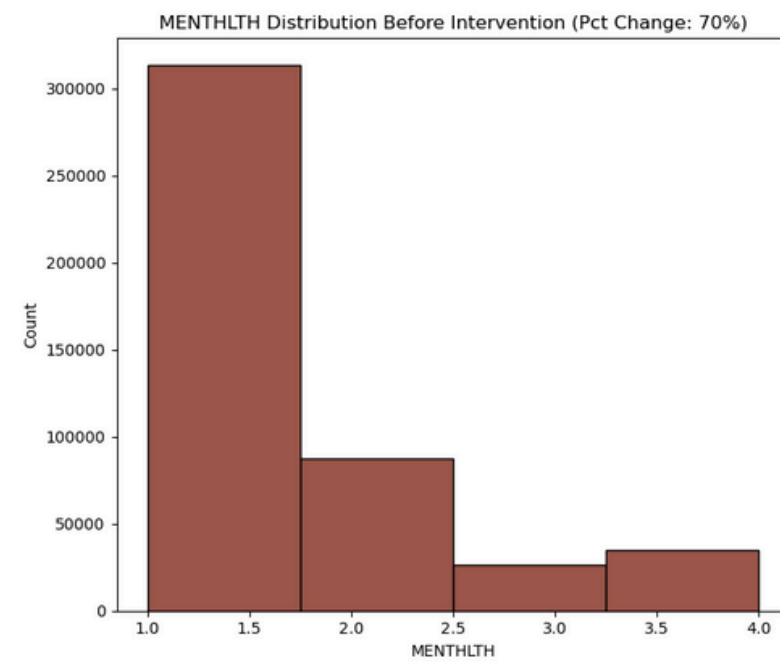
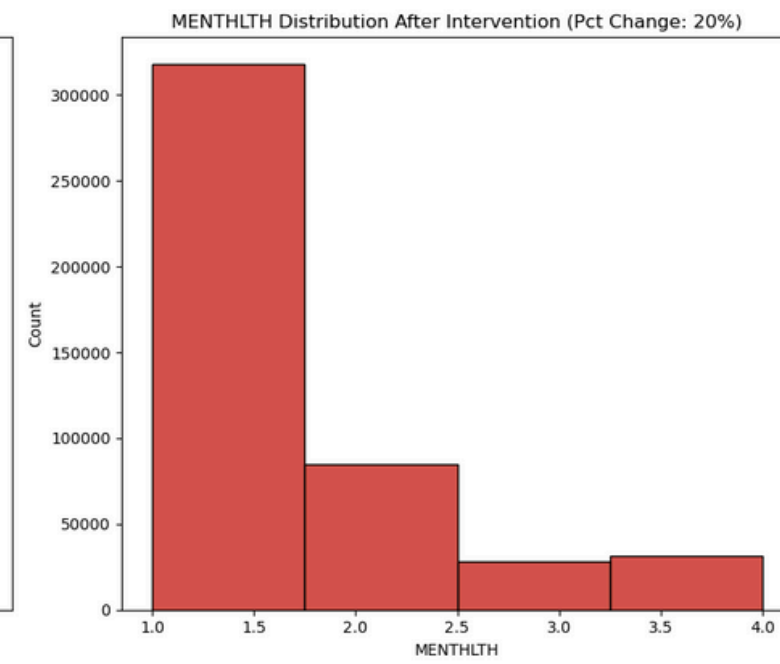
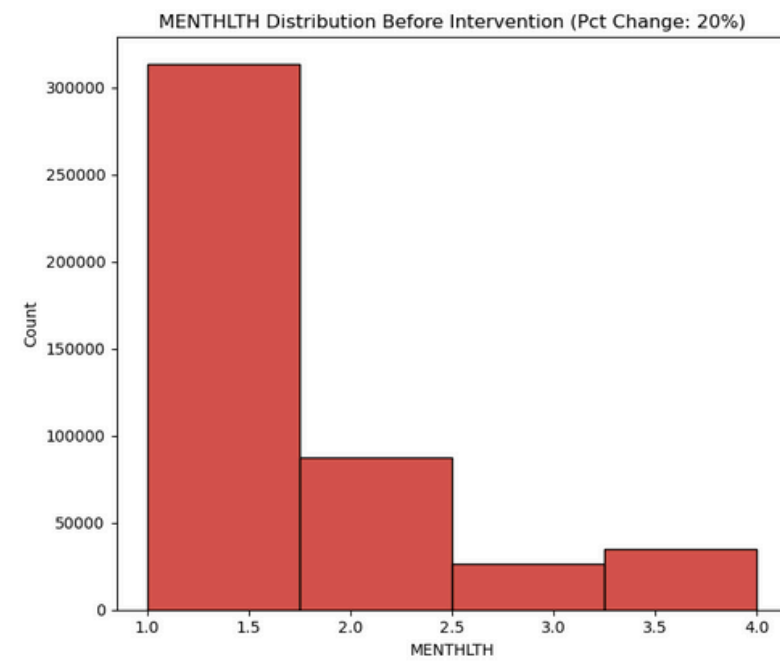
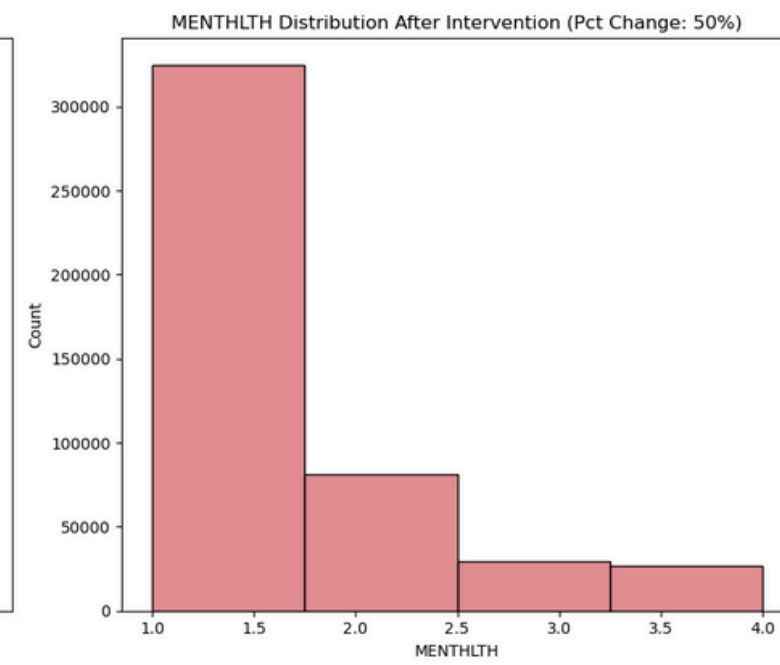
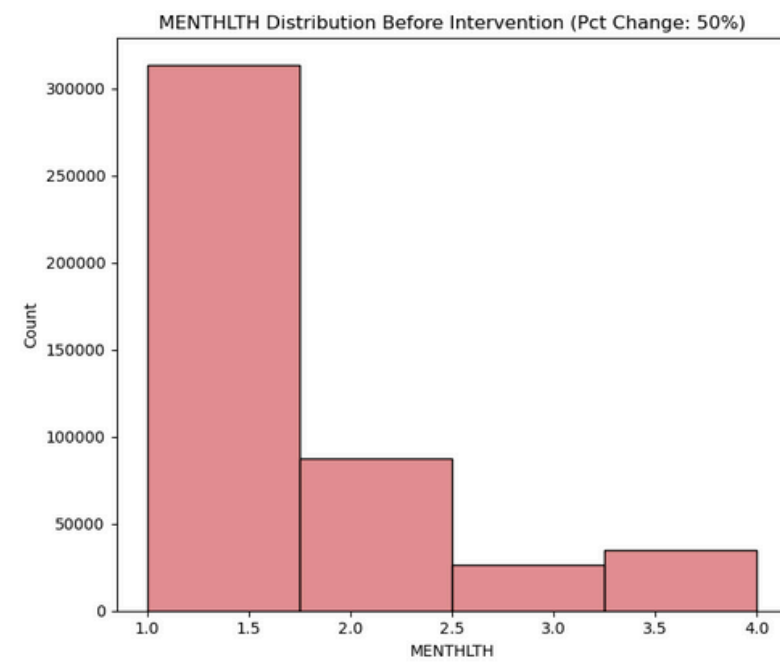
We also perform sensitivity analysis for % change in MENTHLTH by (20%, 50% and 70%) for lowest income groups 1 and 2

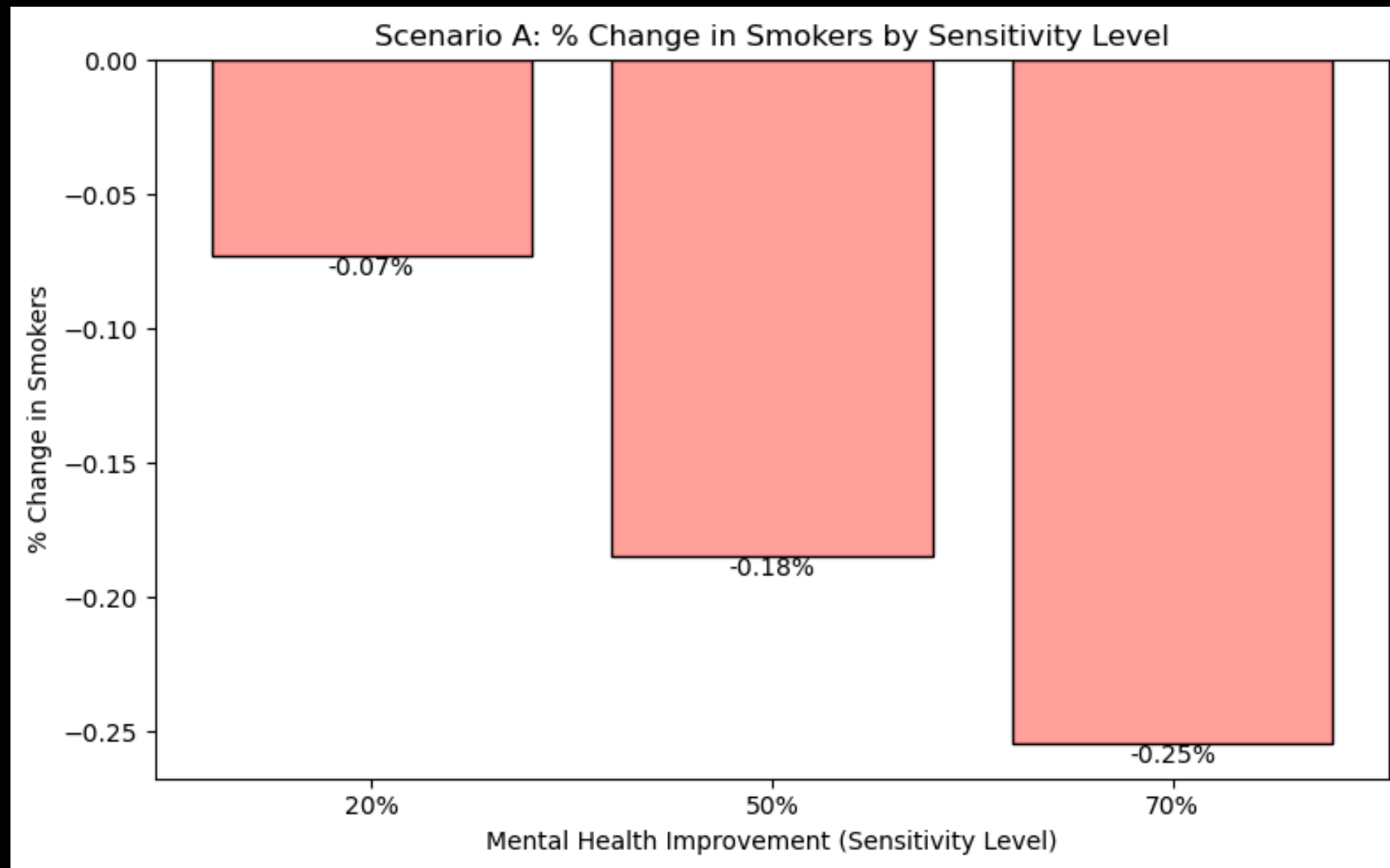
Metrics to report

expected_smokers_base and expected_smokers_sim (absolute counts)

Absolute and percent reduction

Stratified results for _INCOMG groups





Interpretation

Mental health support may play a supportive role but is not sufficient by itself to drive large reductions in smoking.

Scenario B - Education Uplift Program

Goal: Simulate the impact of improving education (health literacy / adult education) in low-income groups.

Rationale: _ EDUCAG showed up to ~29% effect on the income coefficient — biggest lever found.

Intervention rule:

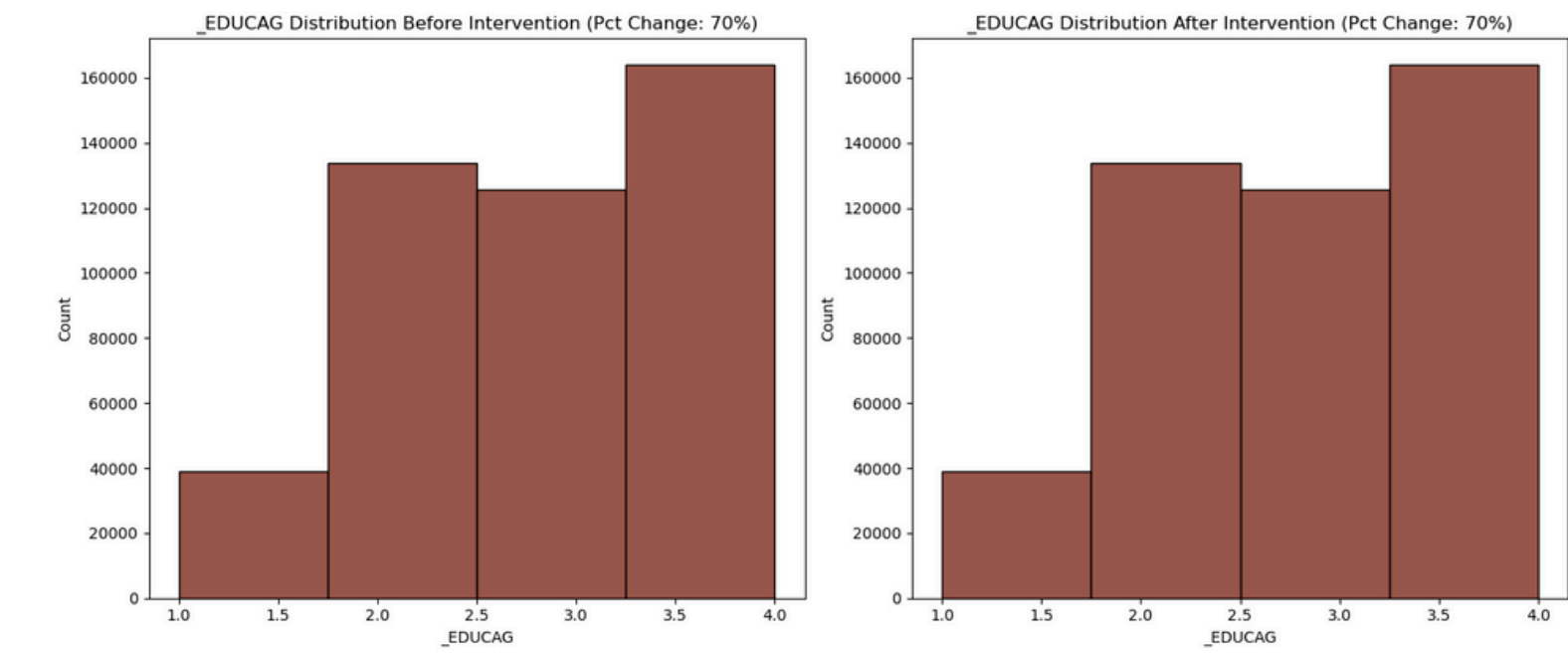
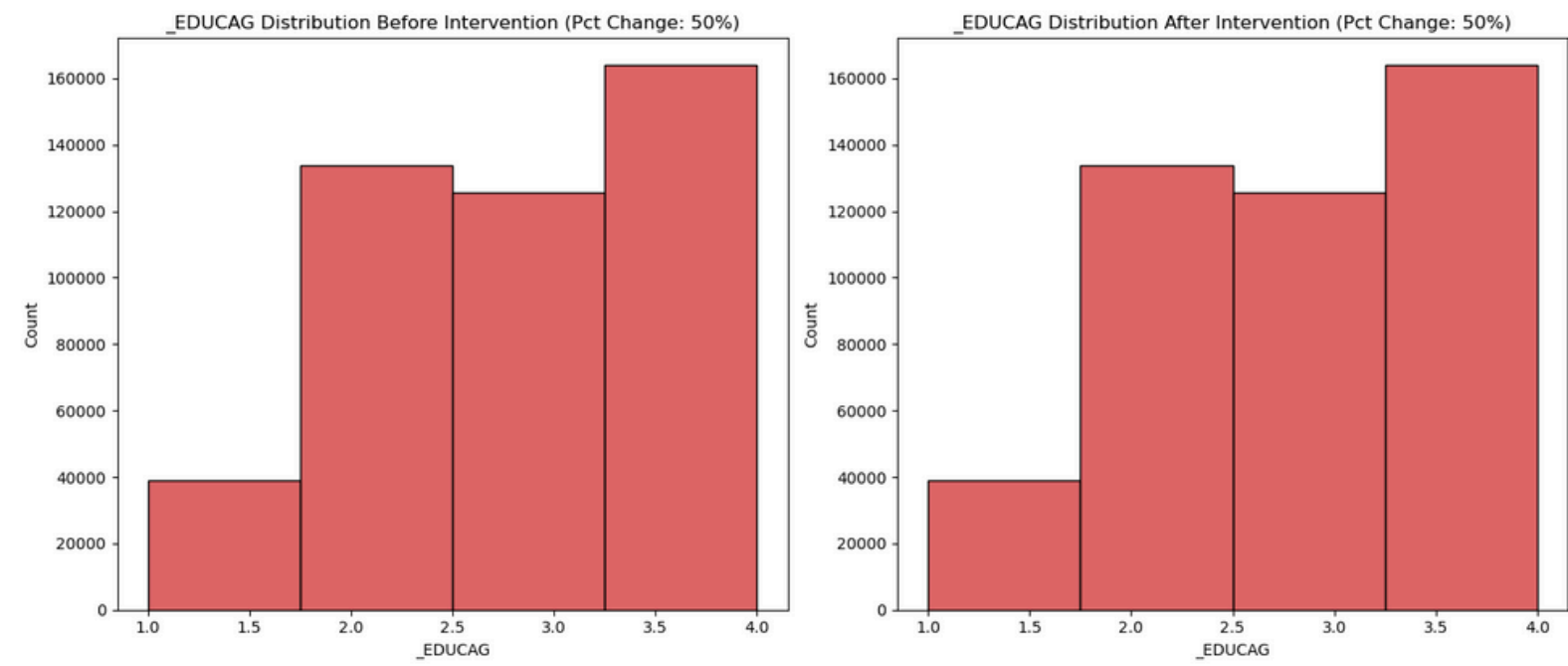
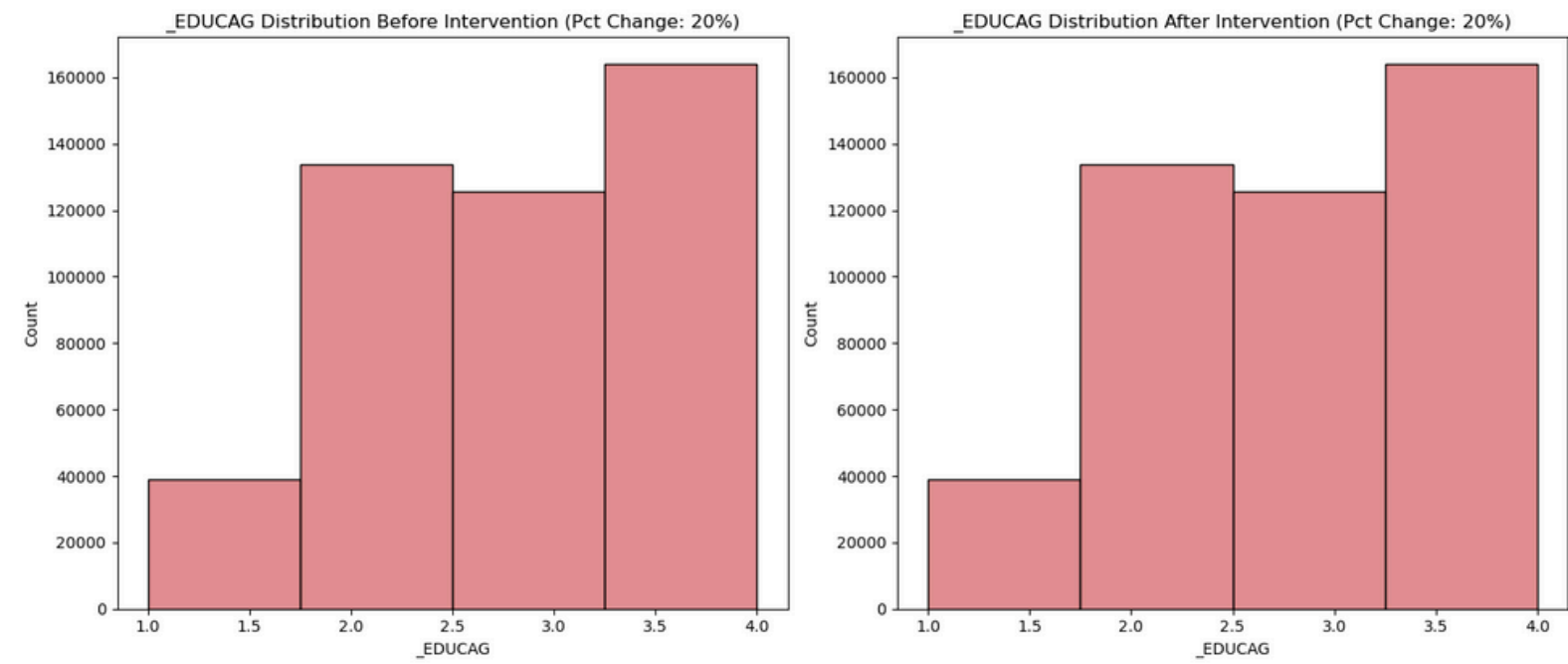
For _INCOMG = 1 & 2, upgrade one education level for 20% of respondents (e.g., Less than HS → High School or HS → Some College). This is a synthetic proxy for adult education / literacy programs.

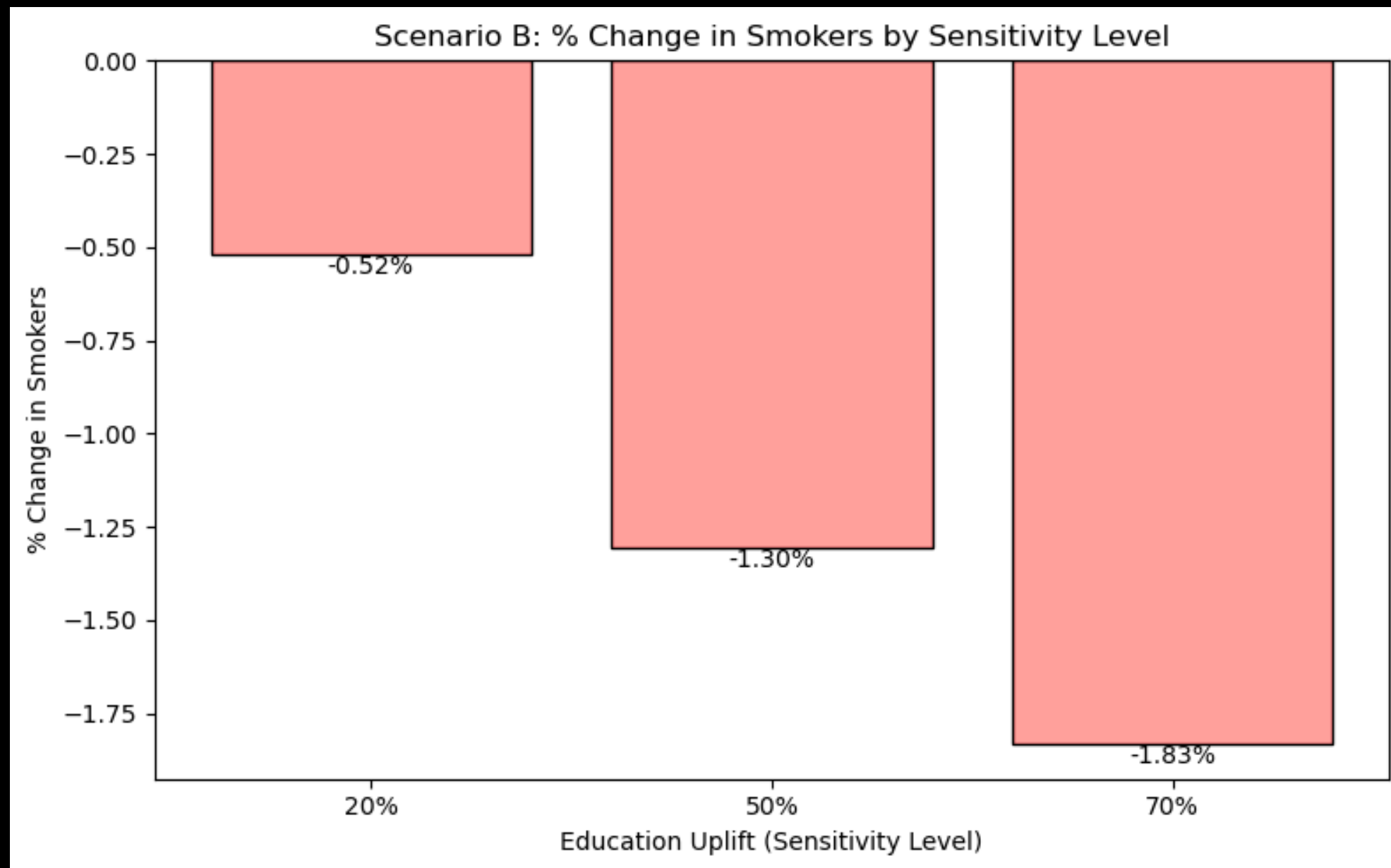
Metrics & checks

Absolute and percent change in expected smokers overall and within _INCOMG groups.

Evaluate distributional effects: does benefit concentrate in certain ages?

Sensitivity: vary upgrade fraction (20%, 50%, 70%) and compare.





Interpretation

As individuals attain higher education, they are more aware of health risks and less likely to engage in smoking.

Scenario C - Remove Medical Cost Barriers + Insurance Outreach

Goal: Reduce smoking by lowering reported medical cost barriers (MEDCOST) and increasing perceived insurance access (HLTHPLN1) in low/mid-income groups. We observe a improvement in GENHLTH which has also been modeled.

Rationale: MEDCOST , HLTHPLN1 and GENHLTH changed the income coefficient 11-21% — an actionable policy lever (subsidies, free cessation treatment).

Intervention rule:

For _INCOMG in [1,2,3],

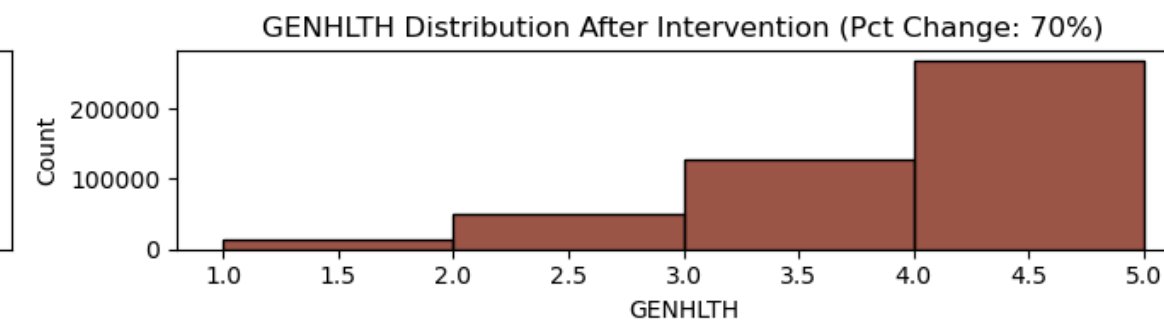
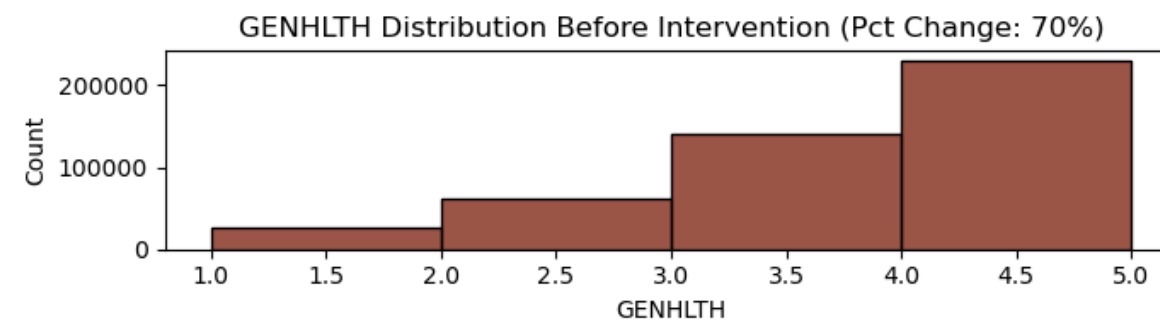
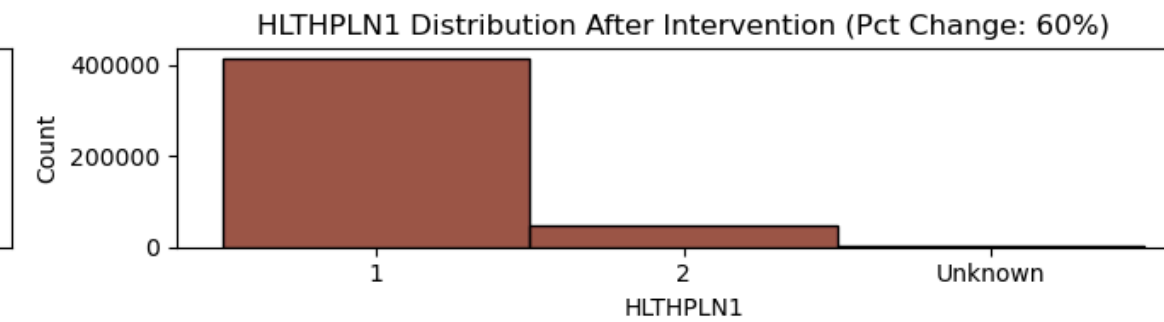
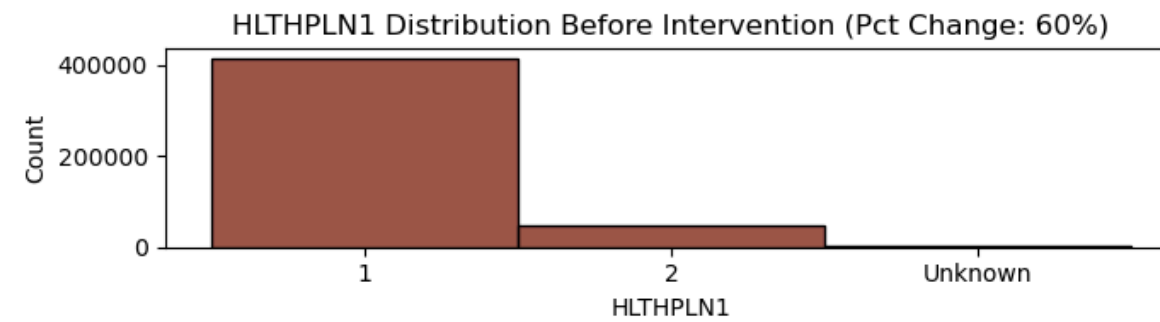
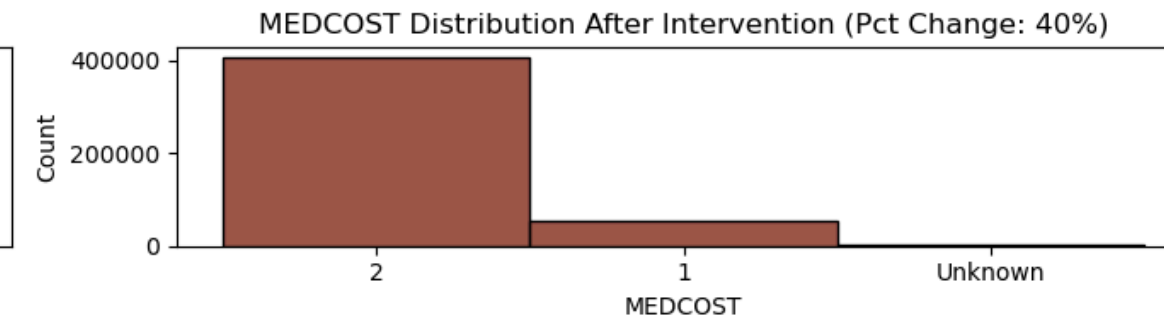
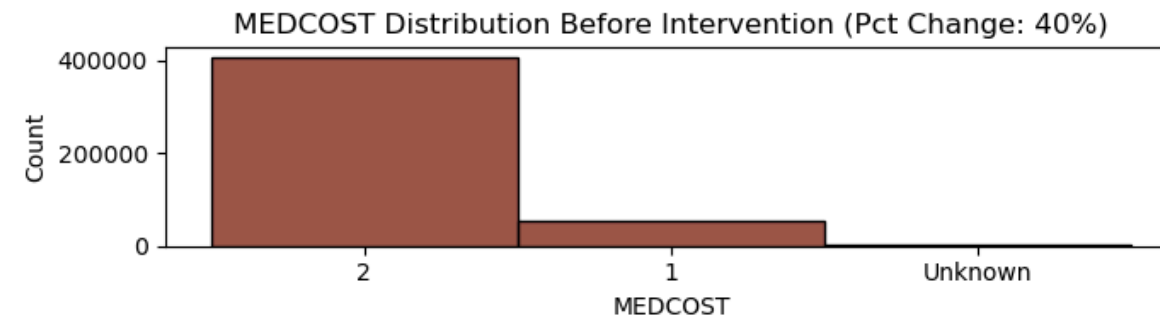
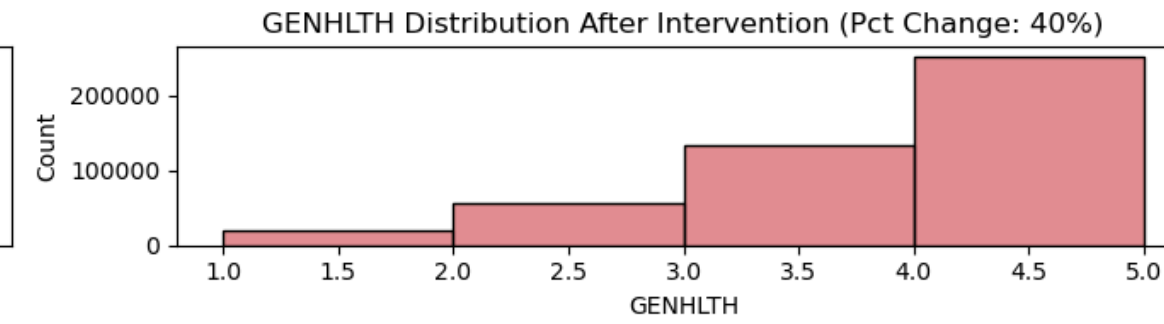
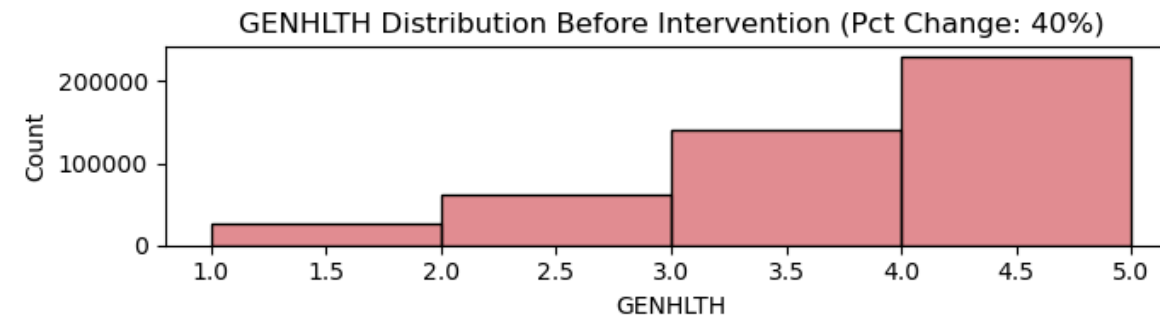
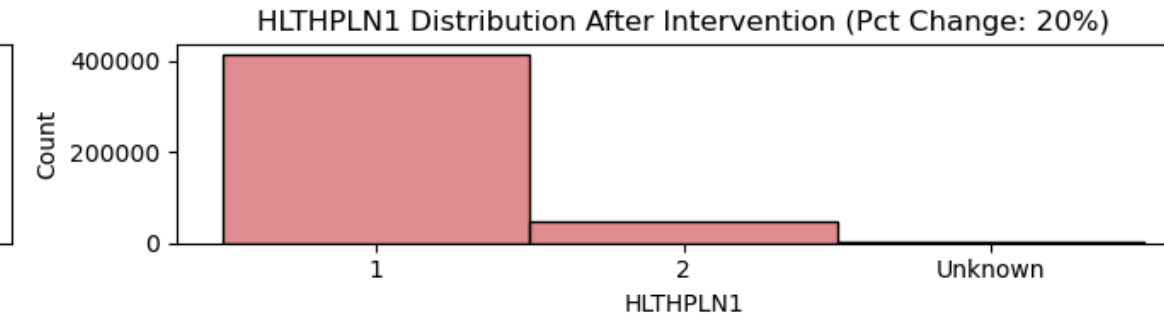
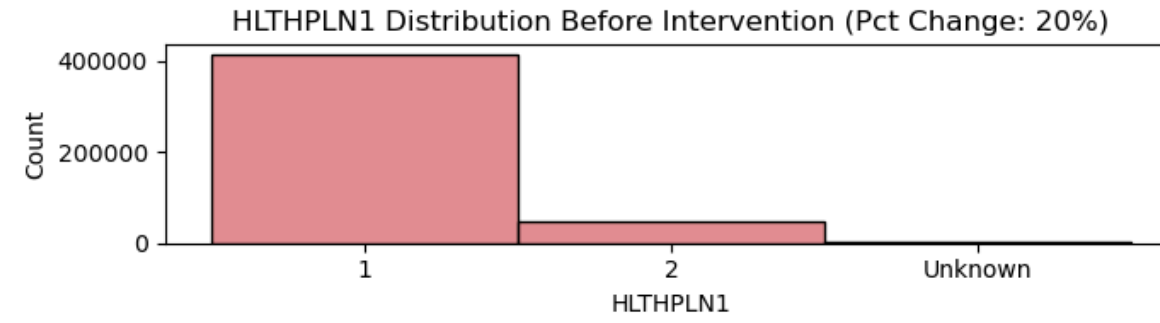
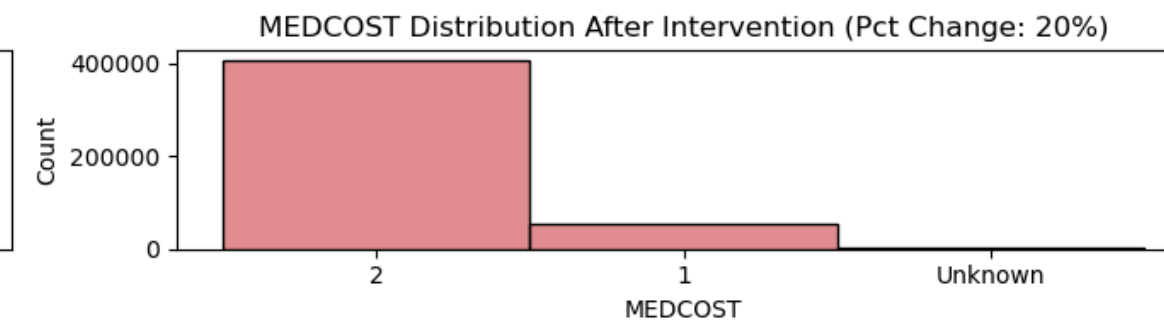
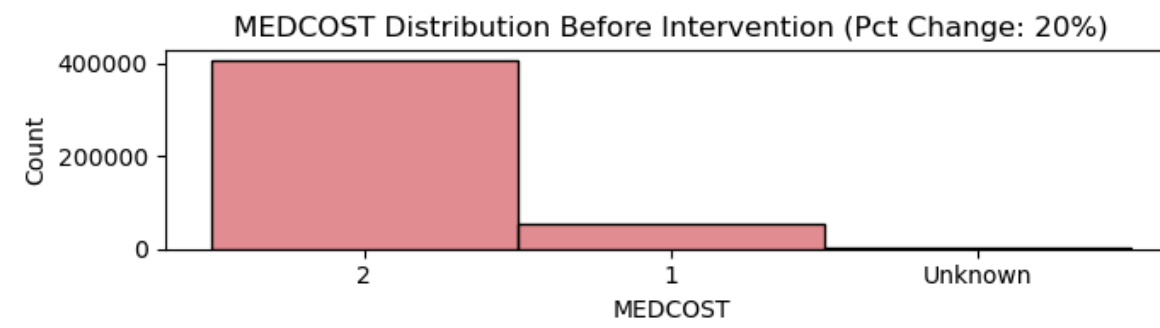
1. Set MEDCOST -> No barrier for 20% of those who reported cost barrier. For same groups, flip HLTHPLN1 to insured for 20% (simulate outreach/coverage). GENHLTH as a result of improved MEDCOST and HLTHPLN1 by 40%. These ratios are picked up from the Confounder Analysis in EDA step.
2. Set MEDCOST -> No barrier for 20% of those who reported cost barrier. For same groups, flip HLTHPLN1 to insured for 30% (simulate outreach/coverage). GENHLTH as a result of improved MEDCOST and HLTHPLN1 by 50%. These ratios are picked up from the Confounder Analysis in EDA step.

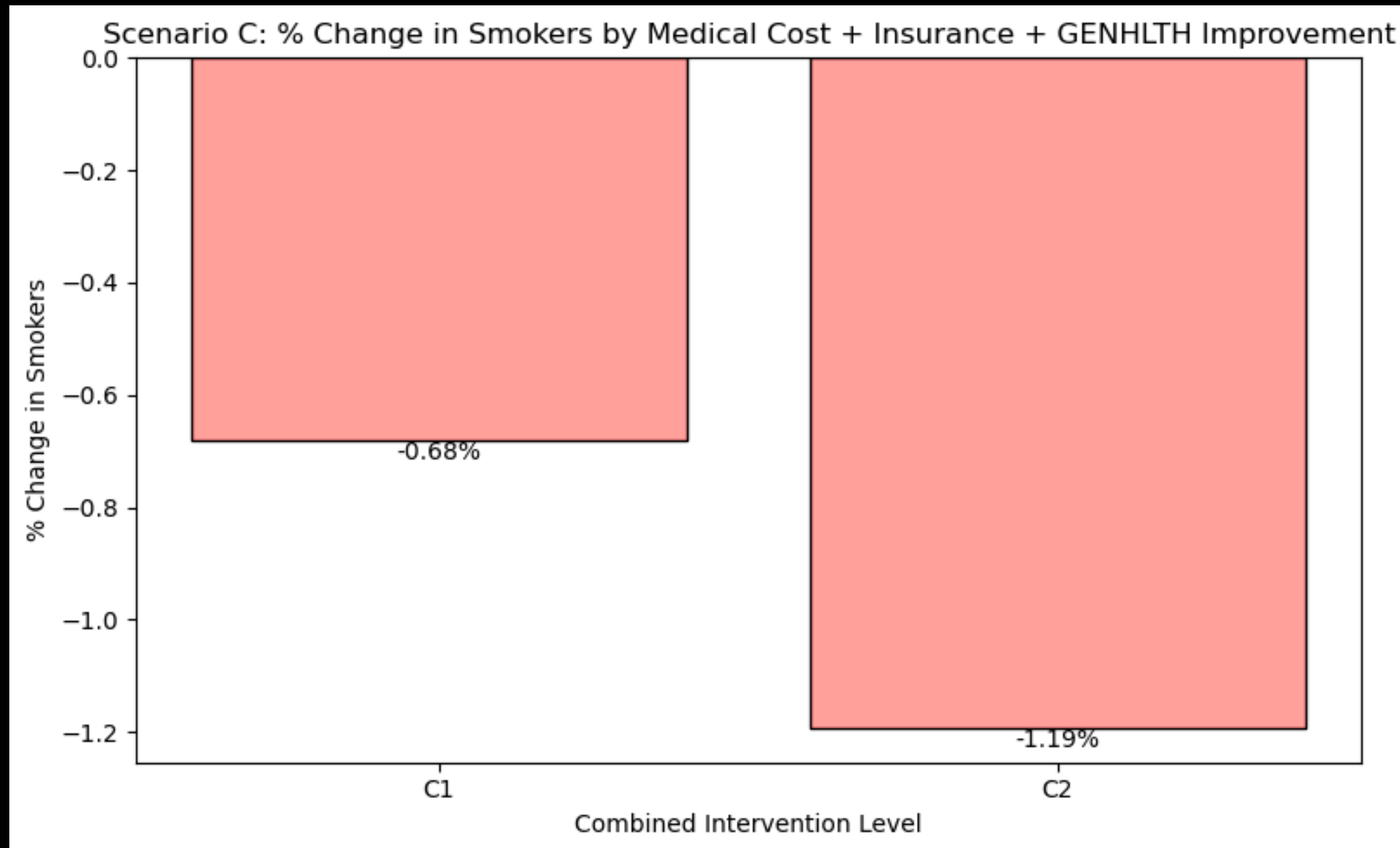
Metrics

Expected smokers baseline vs simulated

Subgroup breakdown by income level and by whether they received benefit

Cost-effectiveness estimate (if you can attach program cost per person)





Interpretation

Reduced medical costs ease financial barriers to care, expanded insurance improves access to cessation programs, and better general health reflects increased well-being and readiness to quit.

